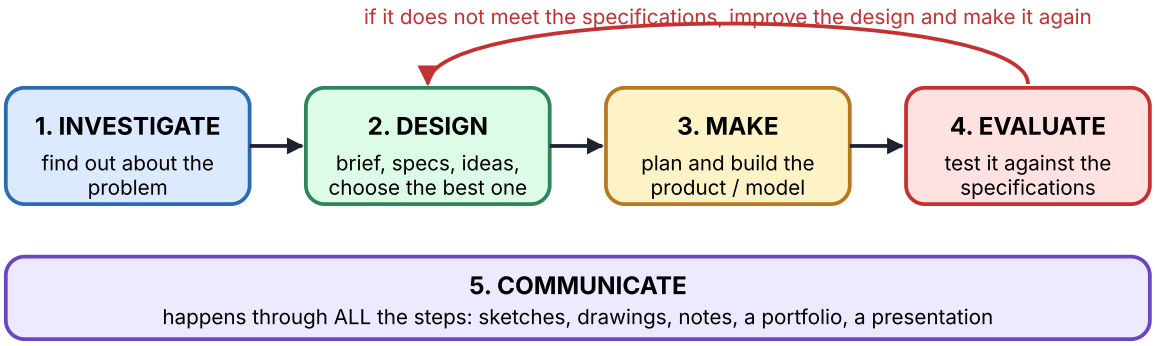
Grade 9 Technology

Easy Study Notes · the short, simple version
Term 1: Technological process · Isometric · Orthographic | Terms 3 & 4: Electrical components + control · Plastics

1. The Technological Process (IDMEC)

The steps you follow to **solve a problem** by designing and making something.



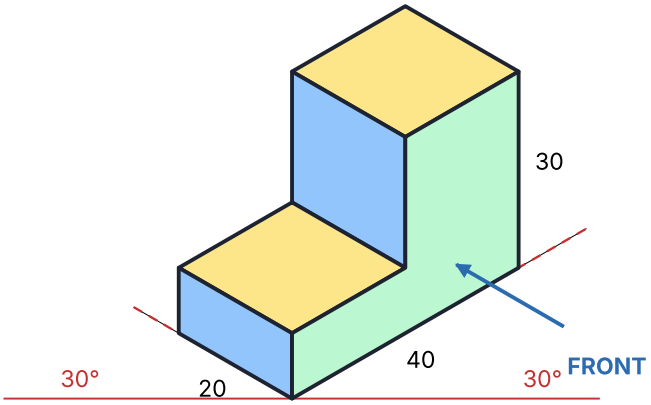
The technological (design) process – IDMEC

Step	In simple words
Investigate	Find out about the problem. Look at products that already exist. Ask: who is it for, what must it do?
Design	Write the design brief and specifications, sketch 2–3 ideas, pick the best one and draw it neatly.
Make	Plan the steps, tools and materials, then build it safely.
Evaluate	Test it: does it meet the specifications? What can be better?
Communicate	Show your work all the way through: sketches, drawings, notes, a portfolio.

Know these 3 words:
Design brief = one sentence: *what* you will make, *for whom* and *why*.
Specifications = what the product **must do / be** (e.g. must be waterproof).
Constraints = the **limits** (e.g. less than R100, finished in 3 weeks).

2. Isometric Drawings

A **3D drawing** – you see the front, side and top at the same time.



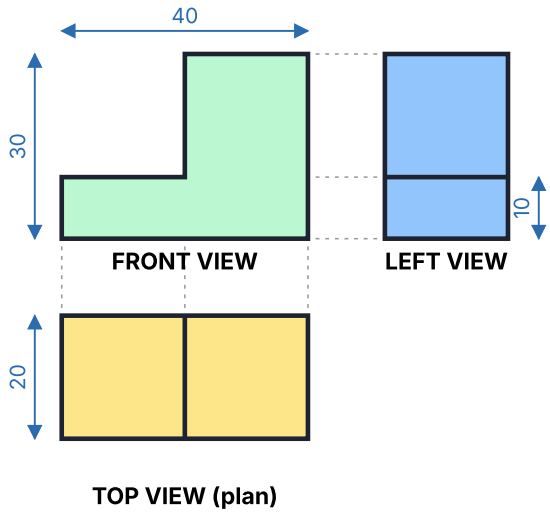
How to read it

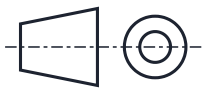
- vertical edges stay vertical
- width and depth edges are drawn at 30°
- lengths along the 3 axes are measured true size
- no hidden (dashed) lines
- 3 faces seen: front, side and top

- **Vertical lines** stay vertical.
- **Horizontal lines** are drawn at **30°** (use a 30°/60° set square).
- Measure **true lengths** along the 3 axes.
- **No hidden lines.**
- How to draw: **1.** draw the 3 axes → **2.** draw a light box → **3.** cut away the parts that aren't there and darken the outline.

3. Orthographic Drawings

2D views of an object, each looking **straight at one side**, with exact sizes. SA uses **first-angle projection**.




first-angle symbol

View	Look from …	Put it …
Front view	the front	top left (the main view)
Top view (plan)	above	**under** the front view
Left view	the left	to the **right** of the front view

> **First angle trick:** the view goes on the **opposite side** to where you look from.

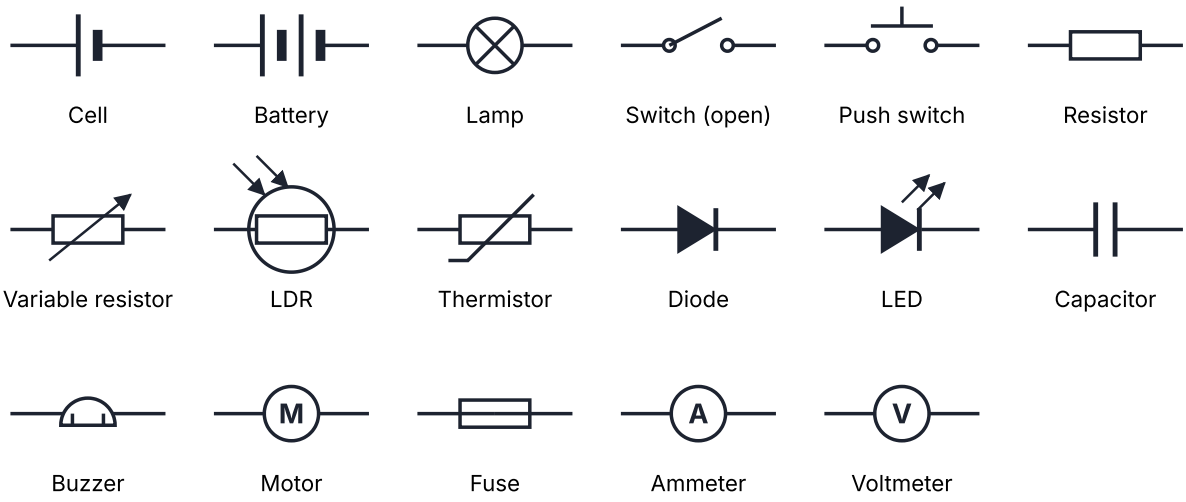
Line	Looks like	Used for
Outline	thick, solid	edges you can see
Hidden line	thin dashes - - - -	edges you can't see
Centre line	long–short chain	middle of holes / circles
Dimension line	thin, arrows on both ends	shows sizes (in **mm**, number written above the line)

Isometric vs orthographic: isometric = 1 picture in 3D, 30° lines, shows what it looks like. Orthographic = several flat 2D views, 90° lines, shows exact sizes for making it.

4. Electrical Components + Control

A **circuit** is a **closed path** for current. It needs a **source** (battery), **wires**, a **load** (lamp, LED, motor) and usually a **switch**.

Quantity	Unit	Measured with
Current (I) – the flow of charge	ampere (A)	ammeter – in **series**
Voltage (V) – the "push"	volt (V)	voltmeter – in **parallel**
Resistance (R) – how much it slows current	ohm (Ω)	ohmmeter / multimeter



Circuit symbols to learn

	Series	Parallel
Paths	one path	more than one path
Lamps	more lamps = dimmer	stay bright
One lamp breaks	all go off	the others stay on

Ohm's law: V = I × R (I = V ÷ R, R = V ÷ I)
Example: 9 V across 450 Ω → I = 9 ÷ 450 = **0,02 A**

Resistor colour code



1 0 × 100 = 1 000 Ω = 1 kΩ (± 5%)

Reading the colour code

Band 1 = first digit
Band 2 = second digit
Band 3 = multiplier (number of zeros)
Band 4 = tolerance (gold ± 5%, silver ± 10%) – read from the other end

Resistor colour code

Colour	Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Grey	White
Number	0	1	2	3	4	5	6	7	8	9

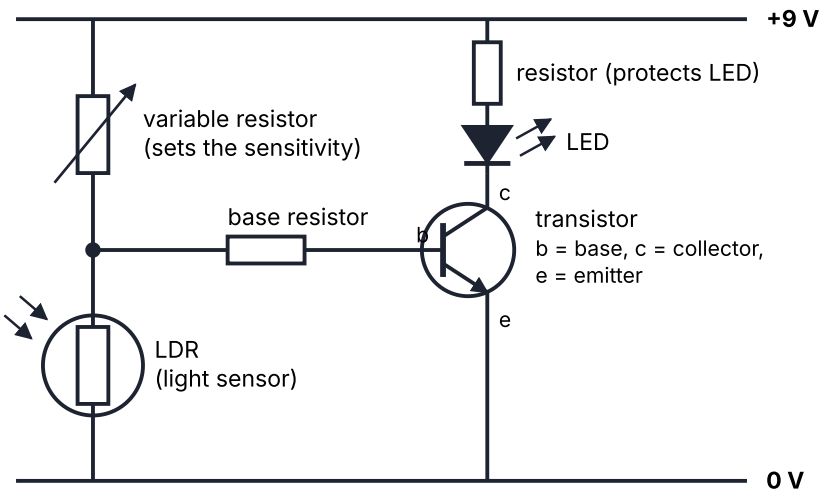
1st band = 1st digit, 2nd band = 2nd digit, 3rd band = number of zeros, 4th band = tolerance (gold ± 5%).

Components – what they do

Component	What it does
Resistor	Limits (reduces) the current. Protects an LED.
Variable resistor	Resistance can be changed with a knob (volume, dimmer).
LDR	Light sensor: **light → low resistance**, **dark → high resistance**. Night lights.
Thermistor	Temperature sensor: **hotter → lower resistance**. Fire alarms.
Diode	Lets current flow **one way only**.
LED	A diode that **gives light**. Needs a resistor in series.
Capacitor	**Stores charge** for a short time – used for time delays.
Transistor	An electronic **switch**: a small current at the **base** switches on a big current (collector → emitter).
Fuse	Melts and breaks the circuit if the current is too big – safety.

Control

Control = deciding **when** an output works. **Systems diagram: INPUT** (switch, LDR) → **PROCESS** (transistor, logic gate) → **OUTPUT** (LED, buzzer, motor).

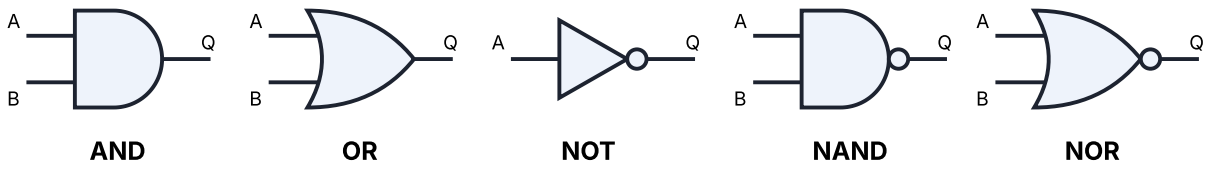


A transistor switch controlled by an LDR: the LED comes on when it gets dark

How the night light works: in the **light** the LDR has low resistance → base voltage too low → transistor **off** → LED off. In the **dark** the LDR has high resistance → base voltage rises → transistor **on** → **LED on**.

Switches: toggle (stays on/off), push-to-make (on only while pushed – doorbell), push-to-break (off while pushed – fridge light), micro switch (microwave door), reed switch (works with a magnet – burglar alarm), tilt switch.

Logic gates



Logic gate symbols (A, B = inputs; Q = output; the small circle means "NOT")

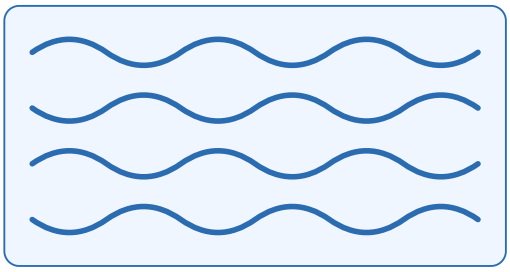
A	B	AND	OR	NAND	NOR		A	NOT
0	0	0	0	1	1		0	1
0	1	0	1	1	0		1	0
1	0	0	1	1	0			
1	1	1	1	0	0			

- **AND** – on only if **both** are on (= switches in **series**).
- **OR** – on if **either one** is on (= switches in **parallel**).
- **NOT** – gives the **opposite**.
- **NAND** = opposite of AND. **NOR** = opposite of OR.

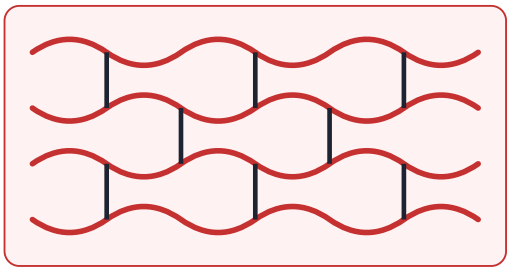
5. Plastics

Plastics are **man-made** materials made from **crude oil**. They are **polymers** – long chains of small repeating units (monomers).

Properties: light, waterproof, don't rust, good **insulators**, easy to shape, cheap, last long.



Thermoplastic: *separate chains with weak forces between them – they slide apart when heated*

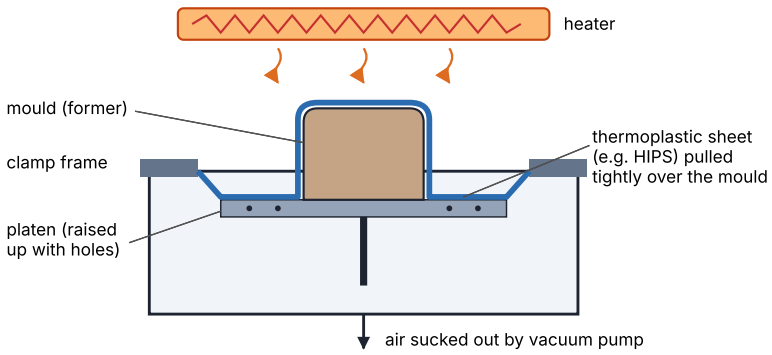


Thermosetting plastic: *chains joined by strong* ***cross-links*** *– a rigid network that can't soften*

	Thermoplastics	Thermosetting plastics
Heat	soften – can be **reshaped again and again**	set hard **once** – can't be reshaped
Recycle?	**yes**	**no**
Examples	acrylic (Perspex), PET bottles, PVC pipes, plastic bags, polystyrene	melamine plates, Bakelite plugs and pot handles, epoxy glue

How plastic products are made

Method	How	Makes
Vacuum forming	Heat a plastic sheet → push the mould up into it → **suck out the air** so the sheet pulls tight over the mould	chocolate box trays, yoghurt pots, packaging
Injection moulding	Melt plastic granules → a screw **forces** it into a closed mould → cool → open	bottle caps, Lego, buckets, chairs
Blow moulding	Hot tube of plastic in a mould → **blow air in** like a balloon	bottles (hollow things)
Extrusion	Push melted plastic through a shaped hole, like toothpaste	pipes, hoses, window frames
Line bending	Heat a sheet along **one line** on a strip heater → bend it → hold until cool	stands, phone holders, signs



Vacuum forming

Plastics and the environment

- Most plastics are **not biodegradable** – they last hundreds of years.
- They pollute land and sea, harm animals and break into **microplastics**.
- Made from **crude oil**, which is non-renewable. Burning them gives off **toxic gases**.
- **Solutions:** Reduce, Reuse, Recycle; biodegradable plastics.
- **Recycling codes** (number 1–7 in a triangle) show the type of plastic so it can be sorted: 1 PET, 2 HDPE, 3 PVC, 4 LDPE, 5 PP, 6 PS, 7 other.